WEEK 1 · 9–12 NOVEMBER 2026

Introduction to hyperspectral Earth observation

From three colours to 426 — what a continuous spectrum buys you

Week 1 · Session 2 · practical: 10_spectral_signatures_and_libraries.ipynb

LEARNING OBJECTIVES

What you will be able to do

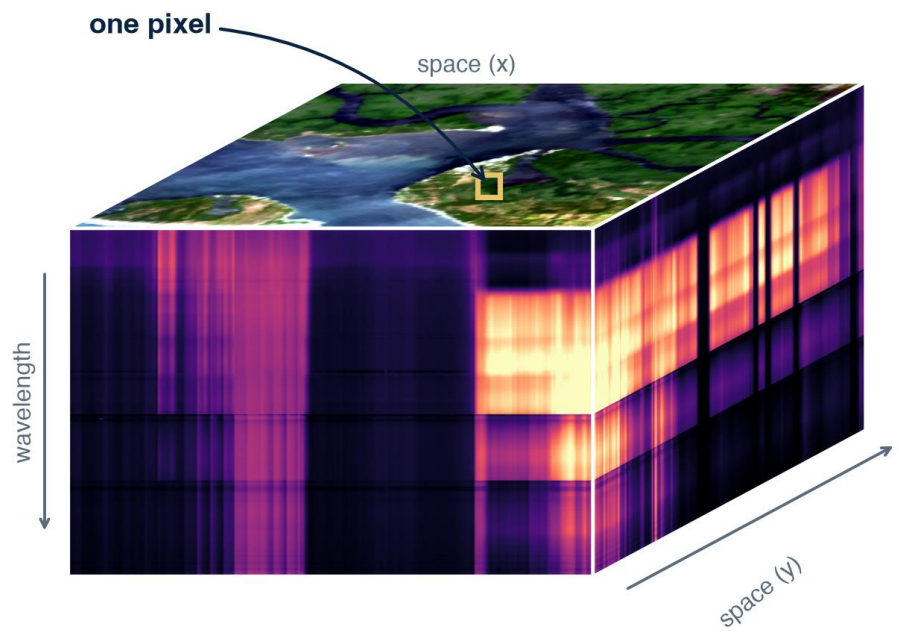
- Explain what makes a sensor *hyperspectral*, and why band count alone is not the definition
- Describe the electromagnetic quantities we measure: radiance, reflectance, emissivity
- Name the main spaceborne imaging spectrometers and their trade-offs
- Say what Tanager-1 measures, at what spectral and spatial sampling
- Judge when hyperspectral data is worth the extra cost and complexity

1. What is hyperspectral?

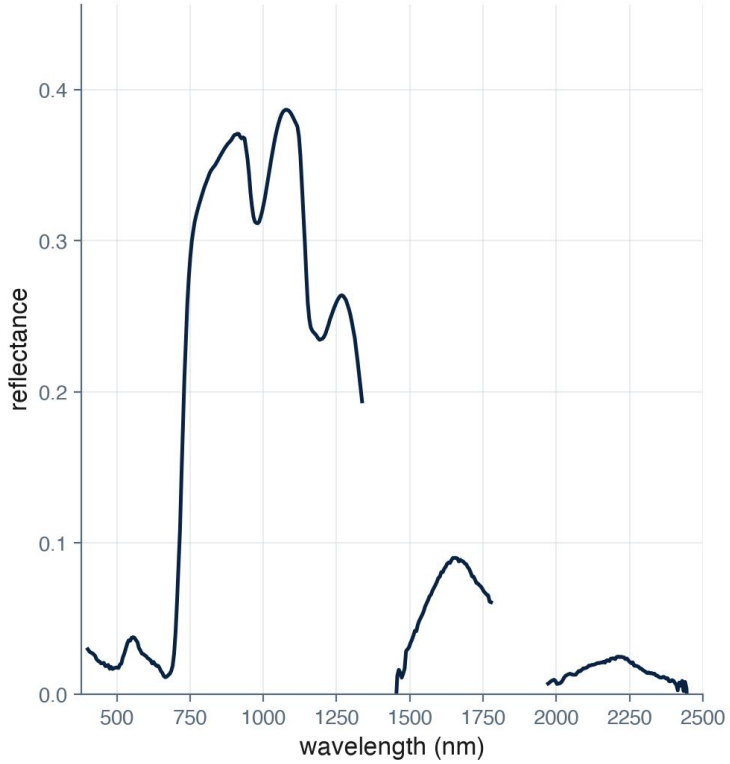
Sampling the spectrum instead of summarising it

You already know the picture. This is what is behind it.

A hyperspectral image is a cube



…holds 426 numbers, not 3



Tanager · Usulután, El Salvador · 352 of 426 bands survive bad-band removal — the two gaps are water vapour

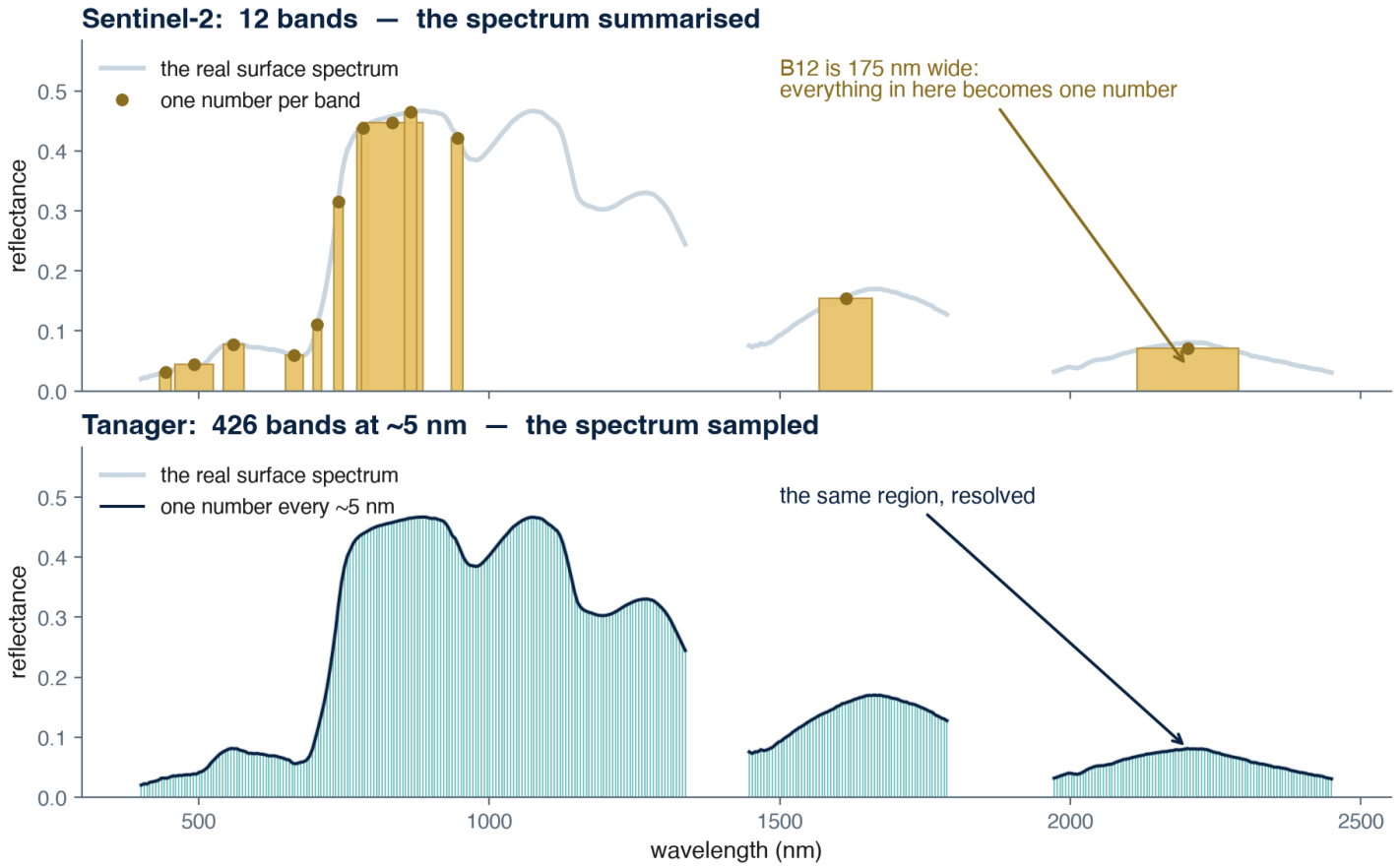
Tanager-1 over Usulután, El Salvador — the top face is the image you would recognise, the depth is wavelength

THE ONE-SENTENCE DEFINITION

A multispectral sensor summarises the spectrum. A hyperspectral sensor samples it.

Sentinel-2 gives you 13 numbers per pixel, each averaging tens to hundreds of nanometres. Tanager gives you 426 numbers, each ~5 nm wide — a curve you can interpret physically.

Summarising versus sampling, on one real spectrum



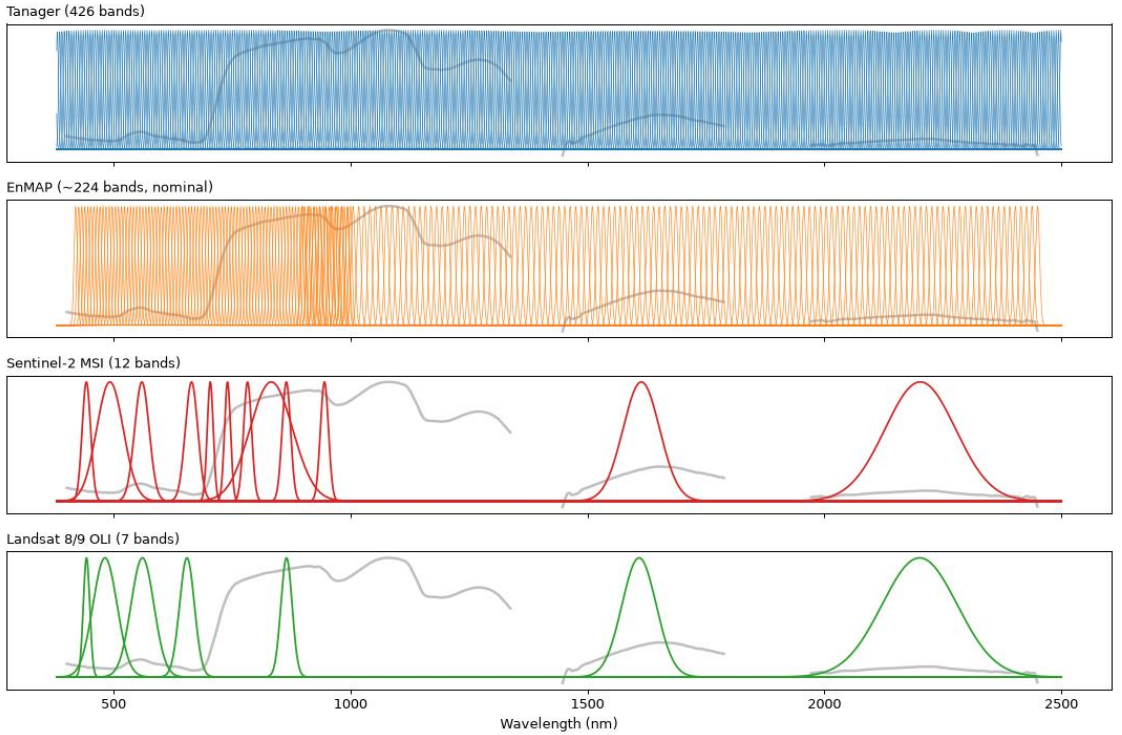
The same irrigated-crop spectrum, as Sentinel-2 records it and as Tanager records it

The three defining properties

- **Contiguous** — no gaps, so the result is a spectrum rather than a set of samples
- **Narrow** — 3–10 nm, enough to resolve a real absorption feature
- **Many** — 100–500 bands, which *follows* from the first two
  - So a 20-band sensor with wide, scattered bands is multispectral, and a 100-band sensor sampling 400–1000 nm continuously is hyperspectral

What the sensors actually see

- Each curve is one band's spectral response function
- Sentinel-2 B12 is ~175 nm wide — it averages the whole SWIR-2 region
- Tanager samples that same region with ~40 bands
- **Features narrower than a band can never be recovered**



Spectral response functions, same wavelength axis

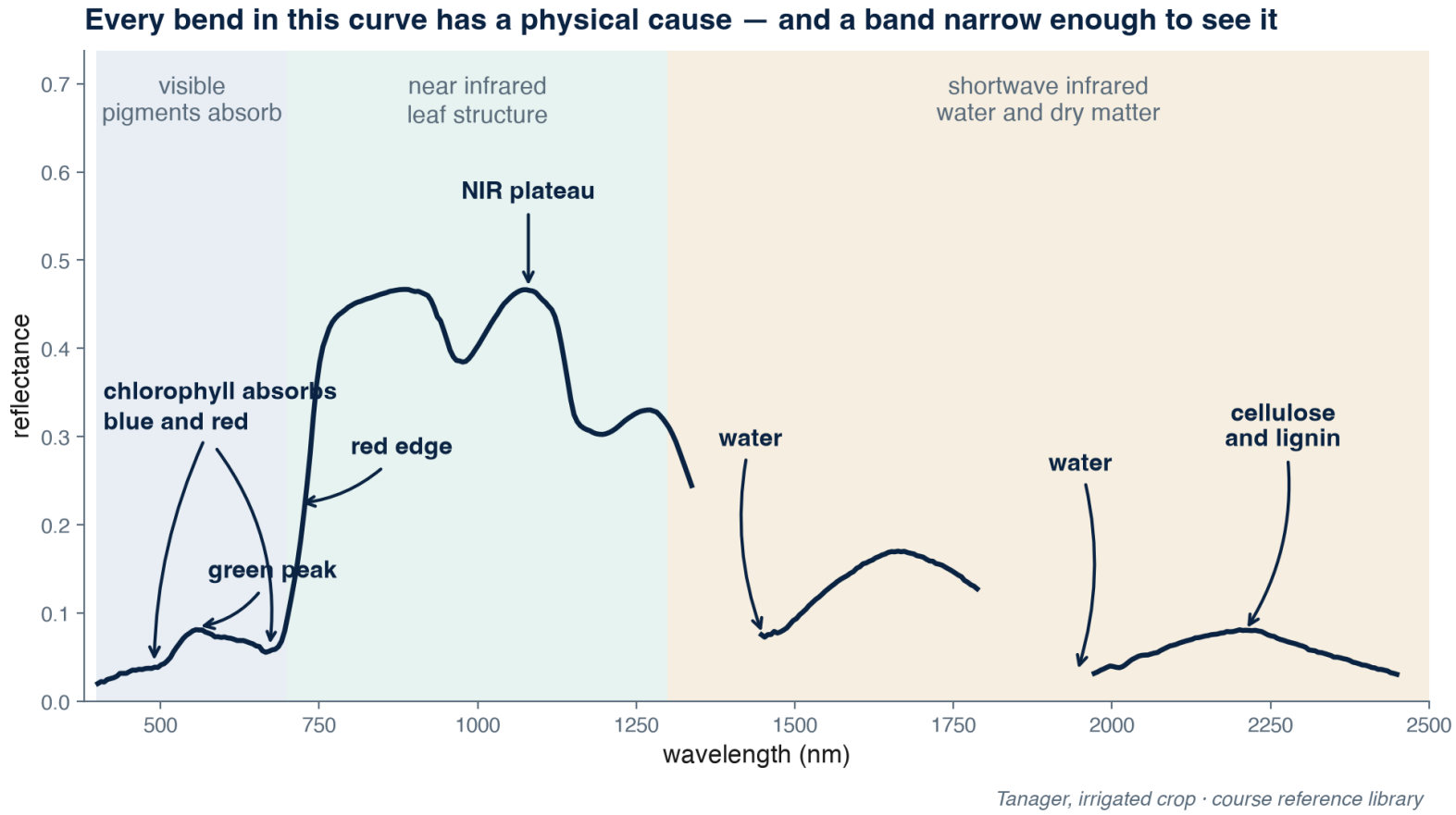
2. What we measure

Radiance, reflectance, and why the difference matters

From photons to a number you can interpret

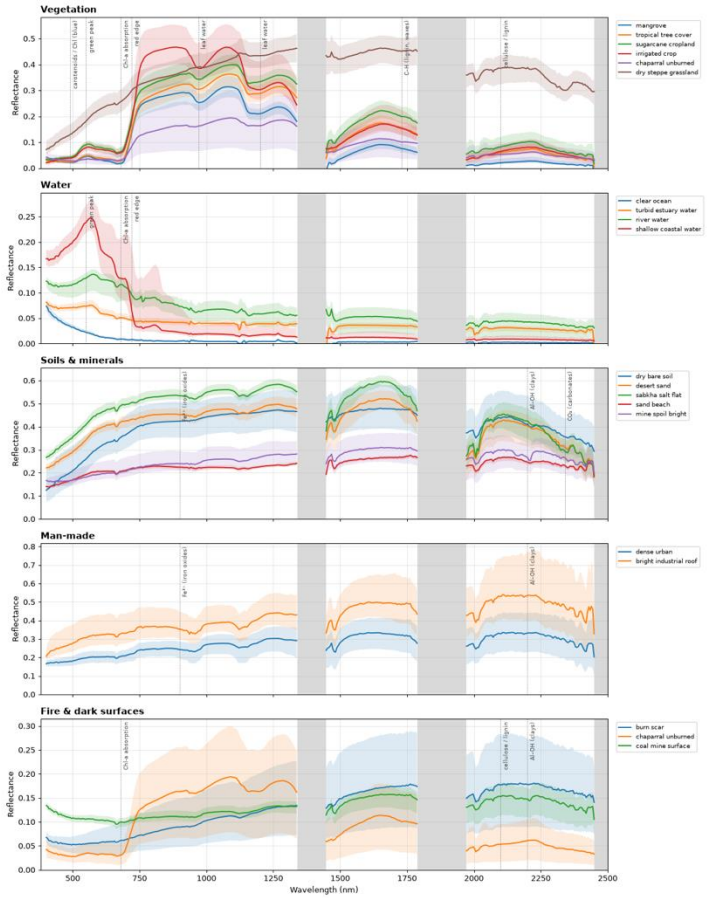
- The sensor measures **at-sensor radiance** — it still contains the atmosphere
- **Atmospheric correction** removes scattering and gas absorption → **surface reflectance**
- Reflectance is unitless (0–1) and comparable between dates, places and sensors
  - Tanager ships both: `ortho_radiance_hdf5` and `ortho_sr_hdf5` — we use the latter, except for gas retrievals, which need radiance

Why a leaf looks the way it does



Pigments in the visible, cell structure in the NIR, water and dry matter in the SWIR

Real Tanager spectra, grouped by material



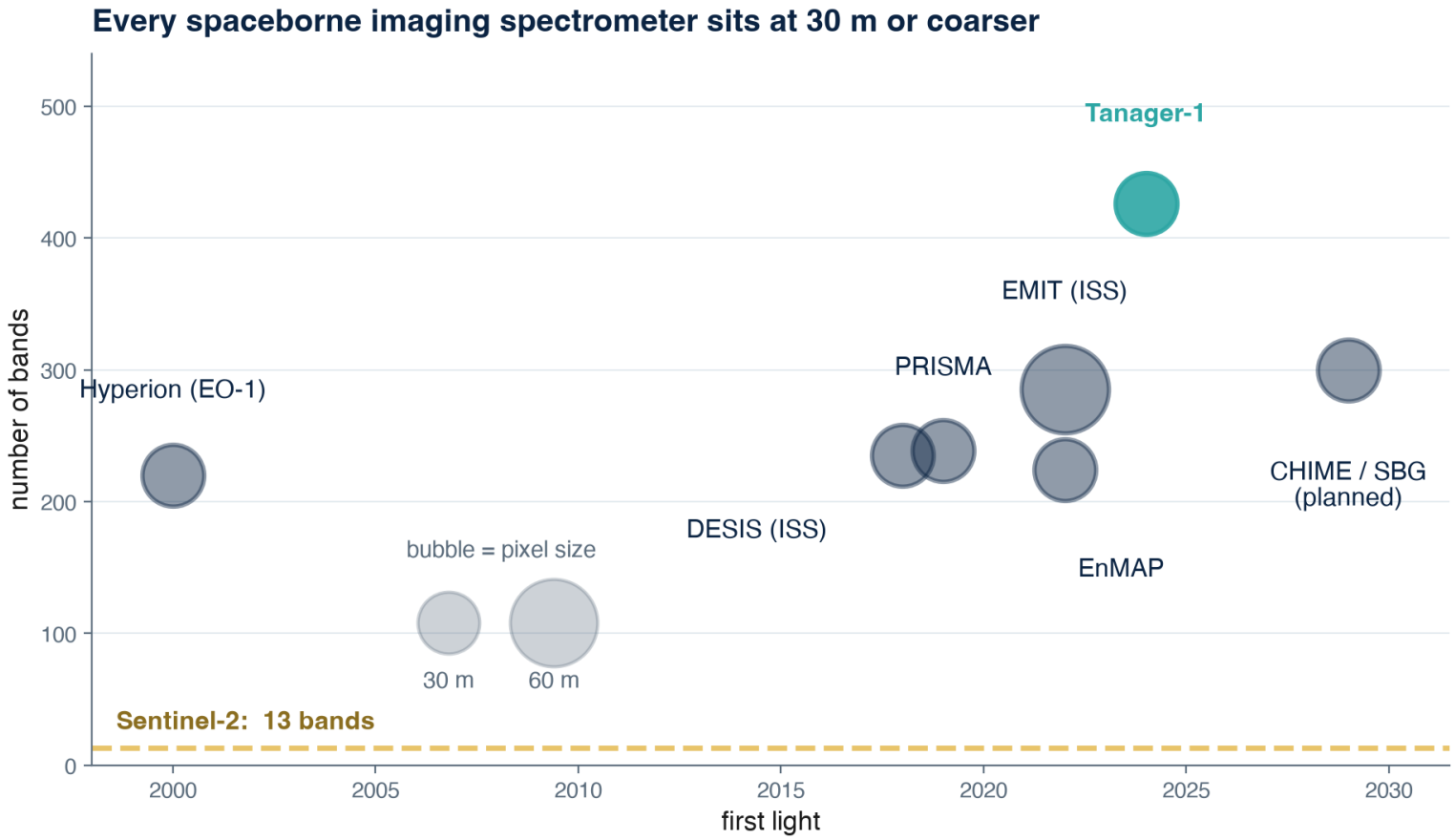
data/spectral_library

- **Measured, not textbook** — every curve comes from the scenes you will analyse
- Vegetation, water, soil, man-made and burnt surfaces, as five families
- The grey columns are the water-vapour windows, where no band survives
- You rebuild this library yourself in the practical
  - Which curve is water, and how can you tell at a glance?

3. The missions

Airborne heritage, today's satellites, what is coming

Where the imaging spectrometers sit



Bubble size is pixel size; the dashed line is Sentinel-2's 13 bands for scale

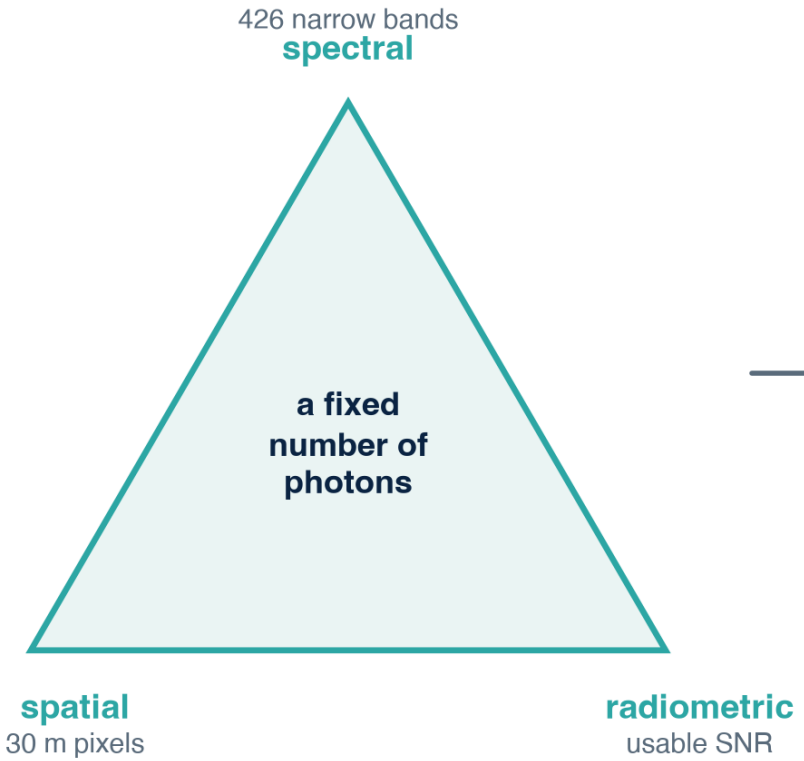
The same missions, with the numbers

Mission	Agency	Since	Bands / range	GSD
Hyperion (EO-1)	NASA	2000–2017	220 · 400–2500 nm	30 m
DESIS (ISS)	DLR/Teledyne	2018	235 · 400–1000 nm	30 m
PRISMA	ASI (Italy)	2019	239 · 400–2500 nm	30 m
EnMAP	DLR (Germany)	2022	224 · 420–2450 nm	30 m
EMIT (ISS)	NASA	2022	285 · 380–2500 nm	60 m
Tanager-1	Planet	2024	426 · 380–2500 nm	30 m
CHIME / SBG	ESA / NASA	planned	continuous VSWIR	~30 m

THE TRADE-OFF

Why 30 m, and not 10 m

Spectral detail is paid for in pixel size and swath



What that buys you

Split the light 426 ways
and each band gets less of it

Keep the pixels large
so every band still beats the noise

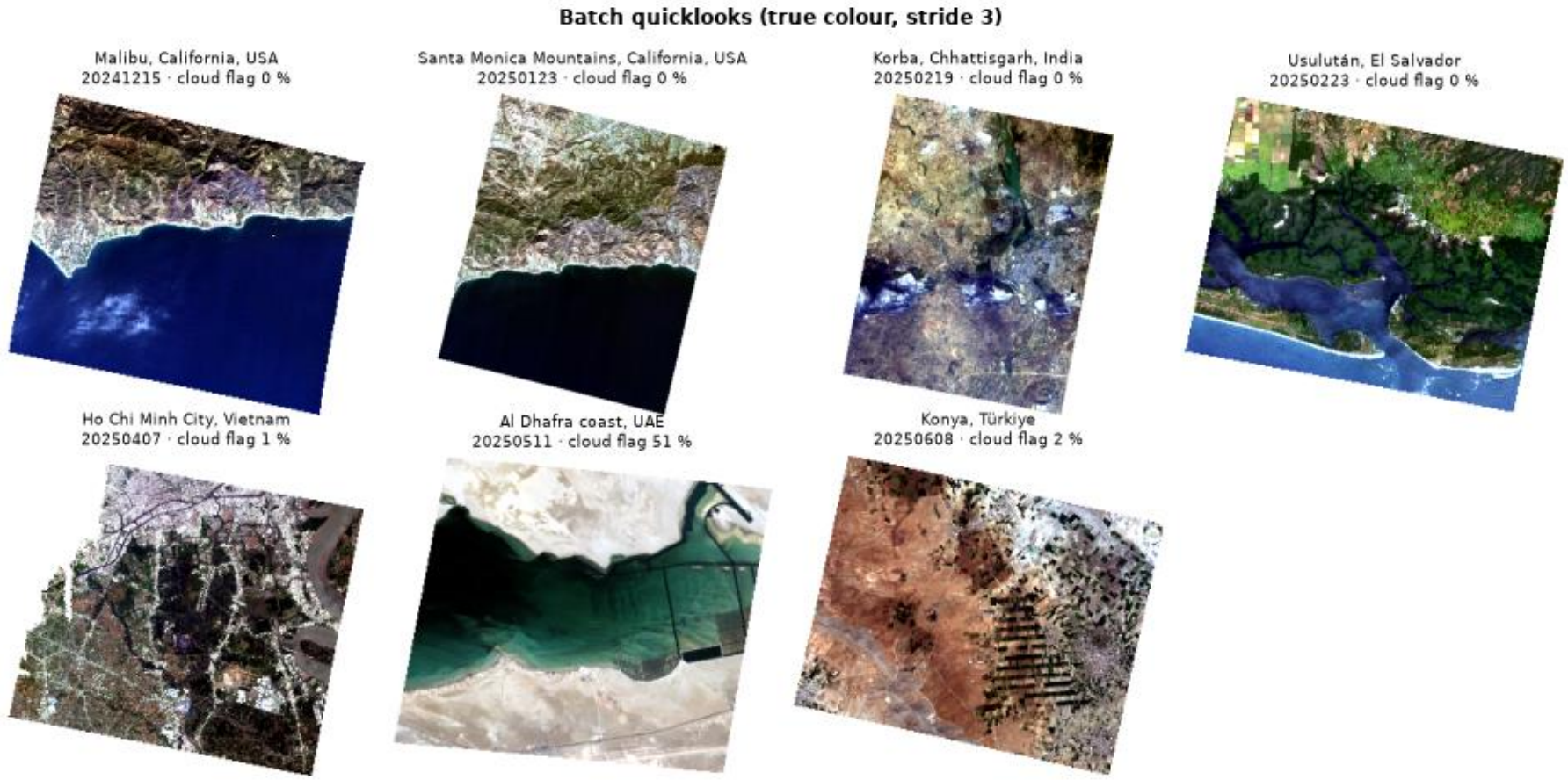
Keep the swath narrow
Tanager sees ~18 km, not 290 km

So: tasked scenes
not wall-to-wall, not daily

Hyperspectral complements Sentinel-2 — it does not replace it

Spectral, spatial and radiometric resolution all draw on the same photons

Tanager-1 open data: the course scenes



Malibu · Santa Monica · Korba · Usulután · Ho Chi Minh City · Al Dhafra · Konya

4. When is it worth it?

Matching the question to the sensor

Does your question need a spectrum?

Question	Multispectral	Hyperspectral
Where is the vegetation?	Yes — NDVI is enough	Overkill
Is this crop water-stressed?	Partly (NDMI)	Better: narrow water bands
Which clay mineral is this?	No	Yes — 2200 nm feature shape
Is there a methane plume?	No	Yes — 2300 nm absorption
Which algal species bloomed?	No	Yes — pigment absorptions
How much crop residue covers the soil?	No	Yes — cellulose at 2100 nm

The six challenge themes, seen spectrally



Each theme has a region of the spectrum that gives the answer away — shaded here

PRACTICAL

Over to you

▶ `10_spectral_signatures_and_libraries.ipynb` · `90 min`

- Plot the course reference library and identify five absorption features by eye
- Compare Tanager, EnMAP, Sentinel-2 and Landsat response functions on the same axis
- Resample real spectra to Sentinel-2 and see which features survive
- Compute spectral-angle separability and find the pair that multispectral data cannot separate

KEY TAKEAWAYS

Recap

- Hyperspectral = **contiguous, narrow** bands — a sampled spectrum, not a summary
- We analyse **surface reflectance**; gas retrievals work on **radiance**
- Absorption features have physical causes: pigments, water, minerals, gases
- Spectral detail is paid for in pixel size and swath — 30 m, tasked, not daily-global
- Use hyperspectral when the answer depends on a **feature's shape**, not just its brightness

Next: Session 3 — Spectral signatures, sensor concepts and use cases (deeper into FWHM, SNR and mixed pixels)

References and further reading

Goetz, A.F.H. et al. (1985). Imaging spectrometry for Earth remote sensing. Science 228, 1147–1153.

Clark, R.N. (1999). Spectroscopy of rocks and minerals, and principles of spectroscopy. In: Manual of Remote Sensing, Vol. 3.

Guanter, L. et al. (2015). The EnMAP spaceborne imaging spectroscopy mission. Remote Sensing 7, 8830–8857.

Thompson, D.R. et al. (2015). Real-time remote detection and measurement for airborne imaging spectroscopy. AMT 8, 4383–4397.

Planet (2024). Tanager-1 open data and STAC catalogue — www.planet.com/data/stac

USGS Spectral Library Version 7 — www.usgs.gov/labs/spectroscopy-lab